

Experiment 19: Naïve Bayes Classification for Bank Loan Prediction

Aim

To implement **Naïve Bayes classification for Bank Loan Prediction** using Python.

Algorithm

Step 1: Import the required libraries.

Step 2: Load the bank loan dataset.

Step 3: Handle missing values.

Step 4: Encode categorical attributes into numerical values.

Step 5: Separate the input features and target variable.

Step 6: Split the dataset into training and testing sets.

Step 7: Create a Gaussian Naïve Bayes classifier.

Step 8: Train the model using the training data.

Step 9: Predict loan approval for the test data.

Step 10: Calculate and display the accuracy.

Program

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import accuracy_score

data = pd.read_csv("bank_loan.csv")

data = data.dropna()

for column in data.select_dtypes(include="object").columns:
    encoder = LabelEncoder()
    data[column] = encoder.fit_transform(data[column])

X = data.iloc[:, :-1]
y = data.iloc[:, -1]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = GaussianNB()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Predictions:", y_pred)
print("Accuracy:", accuracy_score(y_test, y_pred))
```

Output

```
... Predictions: [1 1 1]
Accuracy: 0.3333333333333333
```

Result

Thus, the **Naïve Bayes classification model for Bank Loan Prediction** was successfully implemented, and loan approval predictions were obtained for the test data.